GARBAGE-GUARD

Development of Computer Vision for Creeks Garbage Detection Using Convolutional Neural Network

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The Researchers

CURRICULUM VITAE



Brian Jay Celeste

Second project developer

System architecture, model training and the Android app · JLPT N3



Genesis Walter D. Reyes II

First project developer

Trained the YOLO26n garbage detection model



Jade C. Gunayan

Testing and analysis

Research testing and statistical analysis of the prototype

THESIS ADVISER



Dr. Jonel R. Macalisang

TUP Manila



King Perth G. Pepito

Systems and documentation

Systems analysis, project rules and proofreading



Rodel S. Banogbanog

Fabrication

Builds and assembles the unit · NC II CSS, NC II Electrical Installation



Garbage in Urban Creeks

BACKGROUND OF THE STUDY



STEP 1
Dumping
Waste thrown into drains ends up in creeks

STEP 2
Build-up
Bottles, bags and containers collect where water slows

STEP 3
Blockage
The pile narrows the channel and slows the current

STEP 4
Flooding
Blocked creeks overflow and leave stagnant, foul water

80%
of riverine plastic emissions into the ocean come from **more than 1,000 rivers**

Meijer et al. (2021)

Monitoring is manual
Barangays learn about a build-up by walking the bank: slow, labor-intensive and only periodic

THE QUESTION

Can a camera tell the barangay when a creek needs clearing?



The Research Gap

BACKGROUND OF THE STUDY

Detection and tracking
Nunkhaw & Miyamoto (2024); Li et al. (2024)

Finds floating waste, but no count over time and no alert

Sensor-based alerts
Te et al. (2024); Arante et al. (2025)

Warns on water level, not on what is in the water

Waste monitoring
Balachandran & Talavera (2023)

Measures bins and containers, not moving water

Surveys and satellites
Meijer et al. (2021); Victoriano et al. (2022)

Periodic readings, nothing to act on the same day

GARBAGE-GUARD COMBINES

- **Detector trained on the creek itself**
  runs at the edge, beside the water
- **Counts only inside a drawn area of water**
  the bank, bridge and barrier are ignored
- **One identity per item across frames**
  a drifting bottle is counted once
- **Threshold alert with a saved snapshot**
  on screen, buzzer and phone

The gap

No reviewed system puts these four pieces together for a creek



Objectives

OBJECTIVES OF THE STUDY

GENERAL OBJECTIVE

Develop a computer vision system for creek garbage detection using a convolutional neural network and YOLO26

01

Design the model

A YOLO26n convolutional neural network that detects

- Bottles
- Bags
- Food containers

Precision, recall and mAP@0.5 on a held-out test set

02

Build the prototype

A Raspberry Pi 5 unit that

- Detects inside an operator-drawn area
- Tracks each item between frames
- Counts and alerts at 20 items (default)

Alert on a touchscreen and an Android app

03

Evaluate the system

Using ISO/IEC 25010:2023

- Functional Suitability
- Performance Efficiency
- Usability

Developer tests plus an end-user survey



Scope and Limitations

SCOPE AND LIMITATIONS OF THE STUDY

SCOPE

- YOLO26n on a Raspberry Pi 5
- Three classes: bottles, bags, food containers
- Dataset split 70/10/20, frozen test set, precision of at least 80%
- Count inside the drawn area, alert at 20 items (adjustable)
- Touchscreen dashboard and Android app
- Field deployment at a creek in Barangay 870, Pandacan

LIMITATIONS

- Floating, visible waste only; sunken or hidden items are missed
- No other materials such as metal, glass or organic matter
- Depends on camera placement and daylight
- Monitoring and early warning only; no collection or water-quality analysis



How Garbage-Guard Works

DESIGN

Camera
HWI-B140H, 1280 x 720

Crop
only the drawn area of water

Detect
YOLO26n (NCNN) at 960px

Track
ByteTrack keeps one ID per item

Count
median of the last 15 frames

? DECISION
Count reached the threshold?
20 items by default · re-arms at 15

One alert per build-up

Raised once at the threshold; re-armed only after the count falls to 15

Touchscreen

Alert banner on the live dashboard

Buzzer

Repeating alarm until muted

Android app

Notification with snapshot via Firebase

History

SQLite record and snapshot



HARDWARE Raspberry Pi 5 (8GB) · Waveshare UPS HAT (E) with four 21700 cells · 7-inch touchscreen · 256GB SSD · passive buzzer · USB Wi-Fi adapter

The Prototype

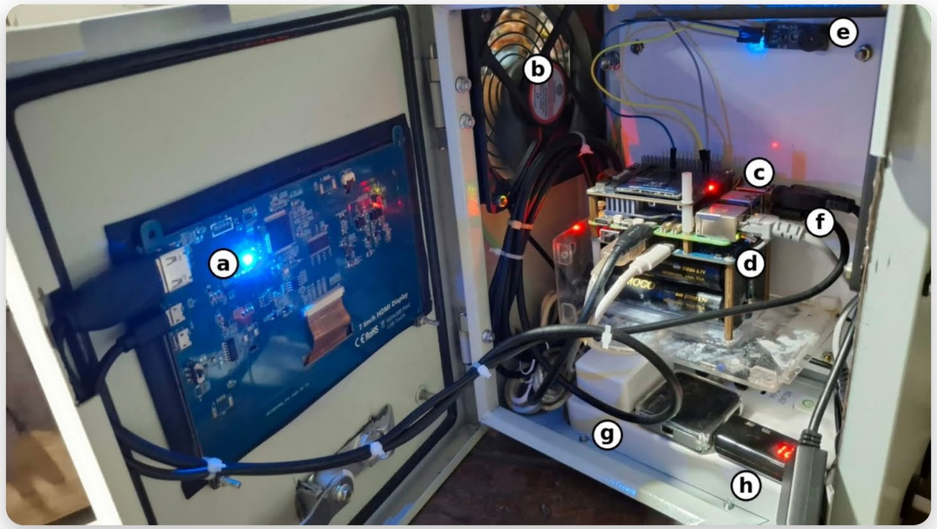
DESIGN/DEVELOPMENT



The assembled unit

Metal junction box, 25 x 30 x 15cm, with the 7-inch touchscreen in the door and a case fan on each side

The camera is mounted outside; only its cable runs to the box



Inside: touchscreen, Pi 5 with SSD and UPS HAT, four 21700 cells, buzzer, camera cable



On site

Camera on the fence above the creek; the UPS keeps it running through power cuts



Dashboard and Mobile App

FEATURES



Touchscreen dashboard during an accumulation alert



Phone alert

Item count, limit and snapshot; mute from the notification

Two screens, one record

The app shows the same alert and counts as the touchscreen, locally or through Firebase



Live boxes

class, ID and confidence

Drawn area

tap points on the water

History

every alert with its snapshot

Remote settings

threshold, buzzer, model upload

Development and Testing

TESTING PROCEDURE

DEVELOPMENT: V-MODEL WITH SCRUM SPRINTS

v1	v3	v4	v5	v6	v6b	v7	v8	v9	v10
public data			first creek photos			test set frozen (807)	960px, field frames		session 3 photos

UNIT

Every sprint

Model on the frozen test set, counter logic, latency logging

10 training runs tested

TC_UNIT_01 to 03

SYSTEM

Integrated unit

Detection, alert, latency, flow, dashboard, area, tracking, storage, delivery, threshold

100 trials, 10 per case

TC_SYS_01 to 10

ACCEPTANCE

Field deployment

Classification, alerting and latency; developer indicators; end-user survey

3 days Nov 6, 7 and 9

TC_ACC_01 to 03



Detection Model Results

TEST RESULTS

84.6%

Precision

Good band · meets the 80% target

67.2%

Recall

misses some items, rarely invents one

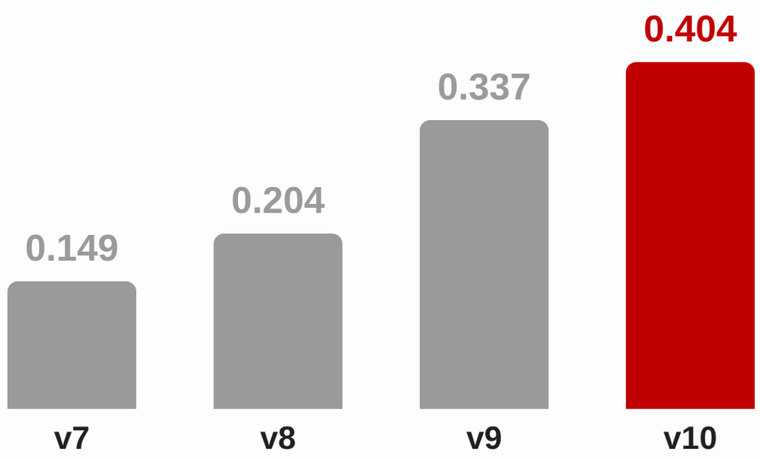
0.744

mAP@0.5

frozen test set of 807 images

Pictures the models never saw

mAP@0.5 on the session 3 test half, 19 pictures, 361 boxes



At the creek

Creek-only precision	68.9%
Food containers (mAP)	0.825
Bottles (mAP)	0.590
Bags (mAP)	0.194

Pictures from the site itself improved the model more than any public dataset



System and Field Results

TEST RESULTS

System tests

TC_SYS_01 to 10, ten trials each

Object detection	10/10
Accumulation alert	10/10
Latency	10/10
System flow	10/10
Live dashboard	10/10
Region of interest	10/10
Tracking and count	10/10
Event storage	10/10
Alert delivery	10/10
Threshold setting	10/10

275-350ms

Average time per frame

Excellent band (below 0.5s)

24 hours

over six days at the creek

No crash, freeze or overheating

0.1-2s

Phone alert on the local network

5s or more through the cloud

19/19

Automated counter tests

Alert once, re-arm at 15

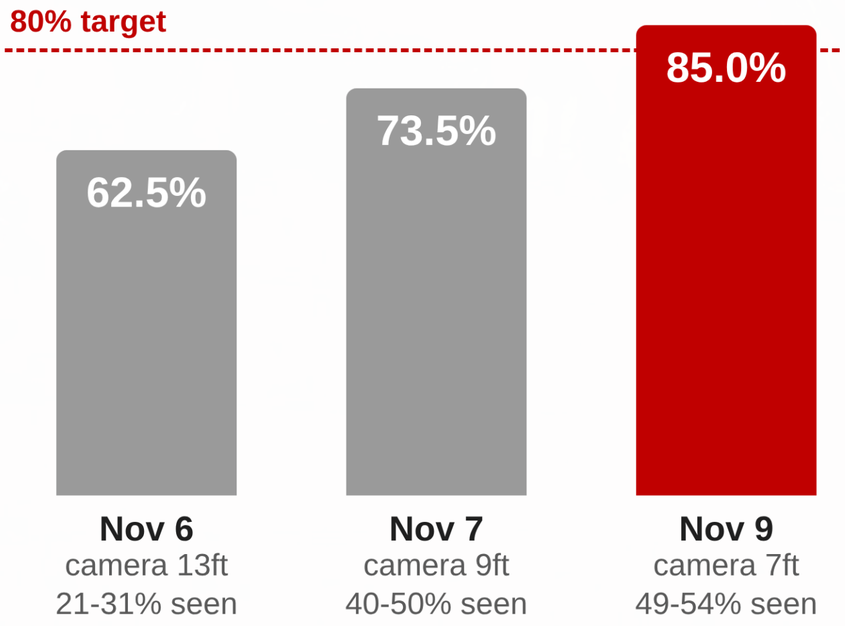


Field Accuracy and Evaluation

TEST RESULTS

Live classification by day

Correct labels over all labels counted · 295 of 395 overall (74.7%)



ISO/IEC 25010:2023

DEV-FS1	Classification	74.7%	Fair
DEV-FS2	Alerting	100%	Excellent
DEV-PE1	Latency	0.28-0.35s	Excellent
DEV-PE2	Stability	24h, no crash	Excellent
USR	Usability	4.40 / 5	Excellent

WHAT CHANGED

Each time the camera went lower, 13ft to 9ft to 7ft, accuracy rose, up to 85.0% at 7ft



Conclusions and Recommendations

CONCLUSIONS AND RECOMMENDATIONS

CONCLUSIONS

1. **Objective 1 met**
   84.6% precision on the frozen test set, above the 80% target

2. **Objective 2 met**
   One Raspberry Pi 5 detects, tracks, counts, alerts and syncs on battery

3. **Objective 3**
   Excellent for speed, stability and usability; field classification Fair at 74.7%

4. **Calibrate on site**
   The count runs low, so the threshold is set from a hand count

RECOMMENDATIONS

1. **Mount the camera low**
   7ft gave the best accuracy; upstream of culverts

2. **More bag images**
   Bags were the weakest class

3. **Retrain from snapshots**
   The unit already saves clean copies

4. **Night and rainy season**
   Infrared camera, longer deployment

5. **Own connection**
   LTE modem, or LoRa for alerts and counts

6. **More cameras, wireless**
   Several cameras per unit; wireless links

7. **Longer range**
   Higher-quality or zoomable camera

8. **Data analytics**
   Trends from the stored alerts and counts

IN SHORT

A workable early-warning tool for barangays, supporting SDGs 6, 9 and 11



THANK YOU!!

